

STAND-ALONE SYSTEM

ELS100

Compact Optical Preservation Appliance

Compact preservation for edge, departmental and branch environments.

A compact automated optical platform for durable nearline preservation. Pair with a performance tier when Active Archive access is required.

**100**

maximum media

Up to 4

optical drives

65 TB

with 1TB media

160W

typical energy

WHERE IT FITS



For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- Edge, lab, departmental and branch repositories
- Evidence, records, imaging and project collections
- Compact preservation or Active Archive deployments
- Pilot deployments that can scale into the ELS family

Technology changes. Information persists.

ELS100

TECHNICAL OVERVIEW

Core specifications

Underlying model	HMS2070
Cartridges / media	2 cartridges; 50 media each; 100 max
Capacity	16 TB / 32 TB / 65 TB (200GB / 500GB / 1TB media)
Drives	Up to 4
Power	110-230VAC 50-60Hz; 450W peak / 160W typical
Dimensions	9.4 x 20.3 x 25.4 in; 240 x 515 x 645 mm
Max weight	79 lb / 36 kg

INTEGRATION & MANAGEMENT

- Standard S3-compatible object access
- oRain Storage Management tracks object and media location
- Write Strategy governs placement and preservation copies
- Browser workflows support discovery and restore

PRESERVATION & RESILIENCE

- Write-once optical media for immutable preservation
- Optical RAID 11+1 protection and recovery model
- Intelligent power management reduces idle energy
- Designed to reduce repeated disk refresh cycles

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

STAND-ALONE SYSTEM

ELS150

Integrated Active Archive & Preservation Appliance Performance and preservation in one compact system.

ELS150 combines SSD/HDD performance storage with automated dual-sided optical preservation - an integrated platform for Active Archive or preservation-first deployments.

**150**

maximum media

Up to 4

optical drives

30 TB

current 200GB media

550W

maximum power

WHERE IT FITS



For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- Compact integrated Active Archive
- Departmental evidence, records and imaging
- Long-lived data requiring fast local access
- Deployments that need performance plus immutable optical preservation

Technology changes. Information persists.

ELS150

TECHNICAL OVERVIEW

Core specifications

Optical configuration	3 standard magazines; up to 150 discs
Media	Dual-sided optical; up to 200GB per disc
Current max capacity	30 TB
Optical drives	Up to 4; up to 80MB/s concurrent throughput
Performance tier	4 x 3.5 in bays; SSD and/or HDD configuration
Protection	Optical RAID 11+1; one parity disc per 12 discs
Interfaces	Gigabit LAN; SAS expansion; S3-compatible protocol
Power	550W maximum; intelligent disk sleep
Dimensions / weight	331 x 592 x 642 mm; up to 55 kg

INTEGRATION & MANAGEMENT

- Integrated performance and nearline preservation tiers
- Performance copy remains while workload requirements justify it
- Preserved optical copy remains after performance copy is removed
- oRain provides object access and storage management

PRESERVATION & RESILIENCE

- Write-once optical anchors the preserved copy
- Automatic recovery model from optical media
- Electronics MTBF >100,000 hours
- Robotic arm rated >2,000,000 cycles

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

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RACK-MOUNT SYSTEM

ELS300

Rack-Mount Nearline Optical Library

High-density nearline preservation in a 7U footprint.

ELS300 provides automated single-sided optical preservation for rack environments. Configure with a separate performance tier for Active Archive.

**288**

maximum media

Up to 4

optical drives

28.8 TB

100GB media

7U

rack footprint

WHERE IT FITS



For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- Rack-based nearline preservation
- Evidence, records, medical imaging and research data
- Active Archive paired with SSD/HDD performance storage
- Data-center deployments requiring compact optical density

Technology changes. Information persists.

ELS300

TECHNICAL OVERVIEW

Core specifications

Optical configuration	24 standard cassettes; up to 288 discs
Media	Single-sided optical; up to 100GB per disc
Maximum capacity	28.8 TB
Optical drives	Up to 4; up to 80MB/s concurrent throughput
Protection	Optical RAID 11+1; one parity disc per 12 discs
Interfaces	Gigabit LAN; SAS expansion; S3-compatible protocol
Power	550W maximum; intelligent power management
Dimensions / weight	448 x 310 x 930 mm; 7U; 80 kg

INTEGRATION & MANAGEMENT

- Separate performance/cache tier supports Active Archive access
- Nearline optical remains available without manual media handling
- oRain provides object access, location awareness and retrieval
- Integrates with larger ELS systems as preservation requirements grow

PRESERVATION & RESILIENCE

- Write-once optical media protects against in-place modification
- Optical RAID protection and automatic recovery model
- Electronics MTBF >100,000 hours
- Robotic arm rated >2,000,000 cycles

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

RACK-MOUNT SYSTEM

ELS500

Scalable Rack-Mount Optical Library

Scalable nearline preservation for growing repositories.

A 7U automated optical library for preservation and Active Archive architectures. Network-accessible drives allow multiple ELS systems to operate within one oRain-managed environment.

**576**

maximum media

Up to 6

drives

374 TB

with 1TB media

7U

rack footprint

WHERE IT FITS



For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- Departmental and enterprise preservation
- Medical imaging, records, evidence and content libraries
- Active Archive behind SSD/HDD performance storage
- Scale-out optical capacity across multiple ELS systems

Technology changes. Information persists.

ELS500

TECHNICAL OVERVIEW

Core specifications

Underlying model	HDL576
Cartridges / media	48 cartridges; 12 media each; 576 max
Magazines	2; 24 cartridges per magazine
Capacity	93 TB / 187 TB / 374 TB (200GB / 500GB / 1TB)
Drives	Up to 6
Power	110-230VAC 50-60Hz; 450W peak / 200W typical
Dimensions	17.7 x 12.2 x 36 in; 448 x 310 x 930 mm
Rack / max weight	7U; 177 lb / 80 kg

INTEGRATION & MANAGEMENT

- S3-compatible object interface
- Pair with performance storage for Active Archive
- oRain manages objects across multiple networked ELS systems
- Write Strategy supports placement, physical separation and additional copies

PRESERVATION & RESILIENCE

- Write-once optical preservation
- Optical RAID 11+1 protection and recovery
- Low idle-power profile compared with disk-only retention
- Preservation capacity can scale without creating storage silos

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

RACK-MOUNT SYSTEM

ELS1000

High-Density Rack-Mount Optical Library

More preservation density without changing the architecture.

ELS1000 expands nearline optical capacity and drive count in a 10U rack footprint while remaining part of the same oRain-managed preservation environment.

**1,152**

maximum media

Up to 12

drives

747 TB

with 1TB media

10U

rack footprint

WHERE IT FITS



For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- High-density enterprise preservation
- Large imaging, evidence and content repositories
- Active Archive paired with performance storage
- Scale-out environments requiring more drive concurrency

Technology changes. Information persists.

ELS1000

TECHNICAL OVERVIEW

Core specifications

Underlying model	HDL1152
Cartridges / media	96 cartridges; 12 media each; 1,152 max
Magazines	4; 24 cartridges per magazine
Capacity	187 TB / 374 TB / 747 TB (200GB / 500GB / 1TB)
Drives	Up to 12
Power	110-230VAC 50-60Hz; 450W peak / 200W typical
Dimensions	17.7 x 17.5 x 36 in; 448 x 445 x 930 mm
Rack / max weight	10U; 352 lb / 160 kg

INTEGRATION & MANAGEMENT

- S3-compatible object access
- oRain provides unified management across ELS systems
- Performance tier can be sized independently of preservation capacity
- Nearline media remains directly addressable through automated libraries

PRESERVATION & RESILIENCE

- Write-once optical preservation
- Optical RAID 11+1 protection and recovery
- Intelligent power management
- Designed for long-lived information that must remain trustworthy

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

RACK-MOUNT SYSTEM

ELS3600

Enterprise Optical Library

Enterprise-scale nearline preservation with high drive concurrency.

ELS3600 provides dense automated optical preservation for large repositories, with up to 24 drives and scale-out management through oRain.

**3,600**

maximum media

Up to 24

drives

2.336 PB

with 1TB media

20U

rack footprint

WHERE IT FITS



For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- Enterprise preservation and tiered storage
- Large image, video, research and evidence collections
- Active Archive at enterprise scale
- Compliance repositories requiring durable media

Technology changes. Information persists.

ELS3600

TECHNICAL OVERVIEW

Core specifications

Underlying model	HDL3600
Cartridges / media	300 cartridges; 12 media each; 3,600 max
Capacity	584 TB / 1,168 TB / 2,336 TB (200GB / 500GB / 1TB)
Drives	Up to 24
Power	110-230VAC 50-60Hz; 670W peak / 200W typical
Dimensions	17.7 x 35 x 43 in; 448 x 890 x 1100 mm
Rack / max weight	20U; 662 lb / 300 kg

INTEGRATION & MANAGEMENT

- S3-compatible access for applications and workflows
- Networked drives support multi-library scale
- oRain maintains object, media and system location
- Pair with independently sized performance storage for Active Archive

PRESERVATION & RESILIENCE

- Write-once optical preservation
- Optical RAID 11+1 protection and recovery
- Low typical energy for large retained capacity
- Supports durable preservation without recurring disk refresh dependence

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

RACK-MOUNT SYSTEM

ELS4000

High-Capacity Enterprise Optical Library

High-capacity preservation in a 20U enterprise platform.

ELS4000 increases media density while retaining the same automated nearline preservation architecture and unified oRain management model.

**4,032**

maximum media

Up to 12

drives

2.616 PB

with 1TB media

20U

rack footprint

WHERE IT FITS



For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- Enterprise preservation and tiered storage
- High-capacity Active Archive environments
- Large image, video, research and evidence collections
- Compliance and long-term information repositories

Technology changes. Information persists.

ELS4000

TECHNICAL OVERVIEW

Core specifications

Underlying model	HDL3600A
Cartridges / media	336 cartridges; 12 media each; 4,032 max
Magazines	14; 24 cartridges per magazine
Capacity	654 TB / 1,308 TB / 2,616 TB (200GB / 500GB / 1TB)
Drives	Up to 12
Power	110-230VAC 50-60Hz; 670W peak / 200W typical
Dimensions	17.7 x 35 x 43 in; 448 x 890 x 1100 mm
Rack / max weight	20U; 662 lb / 300 kg

INTEGRATION & MANAGEMENT

- S3-compatible object access
- oRain manages scale across multiple ELS systems
- Nearline preservation can coexist with additional offline copies
- Performance storage remains independent and workload-sized

PRESERVATION & RESILIENCE

- Write-once optical preservation
- Optical RAID 11+1 protection and recovery
- Intelligent sleep/power management
- Designed for resilient long-term storage at enterprise scale

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

RACK / FLOOR SYSTEM

ELS8000

Petabyte-Class Nearline Optical Library

Petabyte-class automated preservation for the enterprise.

ELS8000 brings 9,504 media positions and up to 40 drives into a large-scale oRain-managed environment for nearline preservation and Active Archive.

**9,504**

maximum media

Up to 40

drives

6.166 PB

with 1TB media

45U

footprint

WHERE IT FITS



For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- Large-scale Active Archive repositories
- Government, research, healthcare, media and AI datasets
- Petabyte-class nearline preservation
- High-density storage for information that must persist for decades

Technology changes. Information persists.

ELS8000

TECHNICAL OVERVIEW

Core specifications

Underlying model	HDL8640A
Cartridges / media	792 cartridges; 12 media each; 9,504 max
Magazines	22; 36 cartridges per magazine
Capacity	1,542 TB / 3,083 TB / 6,166 TB (200GB / 500GB / 1TB)
Drives	Up to 40
Power	90-300VAC; 1500W peak / 340W typical
Dimensions	23.6 x 78.74 x 42 in; 600 x 2000 x 1067 mm
Rack / max weight	45U; 1,490 lb / 676 kg

INTEGRATION & MANAGEMENT

- S3-compatible object access
- Network-accessible drives support scale across libraries
- oRain knows which system and media hold each object
- Pairs with ELS8000-OL or ELS10K-OL for physically isolated copies

PRESERVATION & RESILIENCE

- Write-once optical preservation
- Optical RAID 11+1 protection and recovery
- High retained capacity at low typical energy
- Supports additional preservation copies and physical separation

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

RACK / FLOOR SYSTEM

ELS10K

Maximum-Density Nearline Optical Library

Maximum-density automated optical preservation.

ELS10K is the highest-density nearline ELS platform, combining 10,368 media positions, up to 40 drives and unified oRain management for very large preservation environments.

**10,368**

maximum media

Up to 40

drives

6.727 PB

with 1TB media

50U

footprint

WHERE IT FITS

For Active Archive, performance storage and nearline preservation work together. For preservation-only deployments, cache can be cleared after successful optical preservation.

BUILT FOR

- Very large Active Archive repositories
- Maximum-density nearline preservation
- Government, research, healthcare, media and AI datasets
- Enterprise environments requiring long-lived, accessible information

Technology changes. Information persists.

ELS10K

TECHNICAL OVERVIEW

Core specifications

Underlying model	HDL10368
Cartridges / media	864 cartridges; 12 media each; 10,368 max
Capacity	1,682 TB / 3,363 TB / 6,727 TB (200GB / 500GB / 1TB)
Drives	Up to 40
Power	110-230VAC 50-60Hz; 1500W peak / 340W typical
Dimensions	23.6 x 86.6 x 47.2 in; 600 x 2200 x 1200 mm
Rack / max weight	50U; 1,546 lb / 701 kg

INTEGRATION & MANAGEMENT

- S3-compatible object access
- Scale across multiple networked ELS libraries
- oRain provides one managed view across performance, nearline and offline tiers
- Pairs with offline ELS systems for isolated preservation copies

PRESERVATION & RESILIENCE

- Write-once optical preservation
- Optical RAID 11+1 protection and recovery
- Low typical energy relative to retained capacity
- Designed for very large long-term information estates

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

ELS systems can operate as preservation infrastructure or as the optical foundation of Active Archive. oRain provides a unified object and storage-management layer across systems.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

OFF-LINE SYSTEM

ELS8000-OL

Offline Optical Preservation Library

Physical isolation without losing management awareness.

ELS8000-OL stores optical media offline for deep preservation and cyber-resilient retention. oRain continues to track the objects, media and physical location while the media is isolated.

**9,504**

maximum media

No drives

media-only

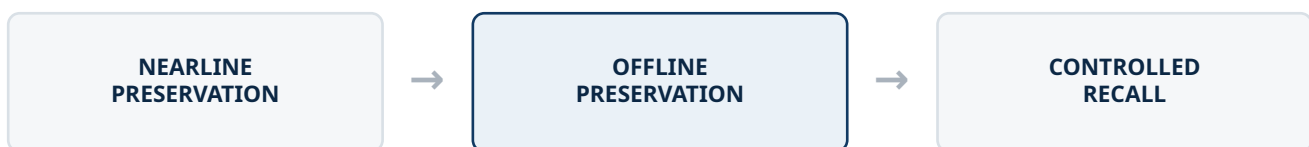
6.166 PB

with 1TB media

45U

footprint

WHERE IT FITS



Offline ELS systems add physical isolation while remaining known to the Savartus management environment.

BUILT FOR

- Offline evidence and records retention
- Physically isolated preservation copies
- Cyber recovery and long-term regulated information
- Deep preservation with indexed recall

Technology changes. Information persists.

ELS8000-OL

TECHNICAL OVERVIEW

Core specifications

Underlying model	HDL8640 OFF
Cartridges / media	792 cartridges; 12 media each; 9,504 max
Capacity	1,542 TB / 3,083 TB / 6,166 TB (200GB / 500GB / 1TB)
Drives	None - media-only library
Power	110-230VAC 50-60Hz; 50W peak / typical
Dimensions	23.6 x 78.74 x 42 in; 600 x 2000 x 1200 mm
Rack / max weight	45U; 1,378 lb / 625 kg

INTEGRATION & MANAGEMENT

- Media remains physically offline until recall is required
- oRain tracks objects, media and physical location
- Required media is moved to an online ELS library for restore
- Can hold an additional isolated copy alongside nearline preservation

PRESERVATION & RESILIENCE

- Physical isolation reduces exposure to online attack paths
- Write-once optical prevents in-place modification
- Low-power deep preservation tier
- Designed for controlled recall rather than continuous access

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

oRain maintains awareness of offline objects and physical media location, so isolated preservation remains part of the managed environment.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.

OFF-LINE SYSTEM

ELS10K-OL

**Maximum-Density Offline
Preservation Library**

**Maximum-density offline
preservation and physical
isolation.**

ELS10K-OL extends the ELS10K media footprint into a media-only offline library for deep preservation, isolated copies and oRain-tracked recall.

**10,368**

maximum media

No drives

media-only

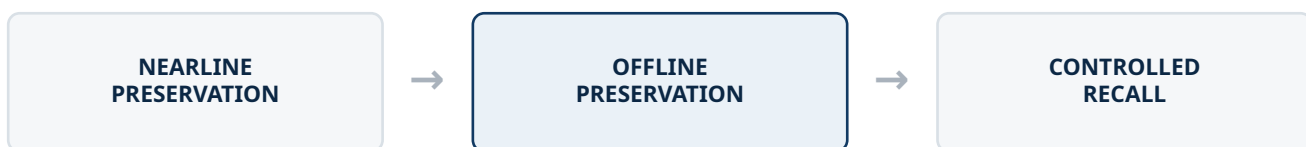
6.727 PB

with 1TB media

50U

footprint

WHERE IT FITS



Offline ELS systems add physical isolation while remaining known to the Savartus management environment.

BUILT FOR

- Maximum-density offline preservation
- Physically isolated enterprise copies
- Cyber recovery and regulated information
- Deep preservation with managed media location

Technology changes. Information persists.

ELS10K-OL

TECHNICAL OVERVIEW

Core specifications

Cartridges / media	864 cartridges; 12 media each; 10,368 max
Magazines	24; 36 cartridges per magazine
Capacity	1,682 TB / 3,363 TB / 6,727 TB (200GB / 500GB / 1TB)
Drives	None - media-only library
Power	110-230VAC 50-60Hz; 50W typical
Dimensions	23.6 x 86.6 x 47.2 in; 600 x 2200 x 1200 mm
Footprint / max weight	50U; 1,546 lb / 701 kg

INTEGRATION & MANAGEMENT

- Same media capacity class as ELS10K, without optical drives
- oRain maintains object and physical media location awareness
- Media is mounted into an online ELS system when restore is required
- Supports additional isolated preservation copies

PRESERVATION & RESILIENCE

- Physical isolation for cyber-resilient preservation
- Write-once optical prevents in-place modification
- Very low active power for retained capacity
- Managed recall keeps offline storage part of one preservation environment

ONE ELS FAMILY. ONE MANAGED ENVIRONMENT.

oRain maintains awareness of offline objects and physical media location, so isolated preservation remains part of the managed environment.

Specifications are configuration-dependent and subject to final system design. Media capacities shown reflect supported media classes in current product specifications.